PCA-BASED ANALYSIS

March 31, 2026

Both dictionary learning and Independent Component Analysis (ICA) presented evidence suggesting that the separator and answer tokens exist in the same subspace, but they do not allow us to determine the basis for the subspace associated with the ICL tasks. This is in part because there is no metric to evaluate the importance of components identified (i.e., independent components and atoms). To address this issue, we repeated our analysis using Principal Component Analysis (PCA). In PCA, principal components are sorted according to the magnitudes of eigenvalues. Further, it is notable that the components with high eigenvalues mediate more information than the ones with the lower eigenvalues. That is, if one can identify a set of principal components with high eigenvalues, which are shared among different types of tokens (query, separator and answer tokens), these principal components could model the subspace underlying ICL.

In this analysis, we analyzed the residual streams of all three types of tokens in ICL prompts (query, separator and answer tokens) by using PCA. Specifically, we collected residual streams from all layers and obtained 30 principal components for each type. That is, PCA components are independently obtained from query, separator and answer tokens. They are referred to as v^Q (query), v^S (separator) and v^A (answer) below.

Fig. 1 shows the cosine distance between 30 principle components of query ($\vec{v_i^Q}$) and answer ($\vec{v_j^A}$) tokens. As shown in the figure, 2-3 principal components of answer and query tokens are similar to each other.

To further test, we collected the cosine distance between the first three principal components (v_1^Q, v_2^Q, v_3^Q) and (v_1^Q, v_2^Q, v_3^Q) of query and answer tokens. Specifically three cosine distances are estimated as in (Eq. 1, 2 and 3).

$$D_1 = D(\vec{v}_1^Q, \vec{v}_1^A), \text{where } D \text{ denotes cosine distance} \tag{1}$$

$$D_2 = D(\vec{v}_2^Q, \vec{v}_2^A) \tag{2}$$

$$D_3 = D(\vec{v}_3^Q, \vec{v}_3^A) \tag{3}$$

As shown in Fig. 2A, D_1 and D_2 are small, but we note that D_3 is high in some cases. This can be explained by the fact that the cosine distance between $\vec{v_1}$ and $\vec{v_2}$ is maximized to 2 when $\vec{v_1} = -\vec{v_2}$. As $\vec{v_1}$ and -$\vec{v_1}$ model the same subspace, we further test the $D' = min(D, 2 - D)$, where D is the cosine distance (Fig. 2B).

We project the residual streams of query, separator and answer tokens in the last layer to subspace spanned by two principal components (Fig. 3). We made three observations. First, all three token types are clustered well according to their types. Second, the query and answer tokens are weakly separated. Third, the separator tokens are separated from other types of tokens. These results raise the possibility that ICL tasks could be solved by vector algebras on the subspace, which map a query cluster onto an answer cluster.

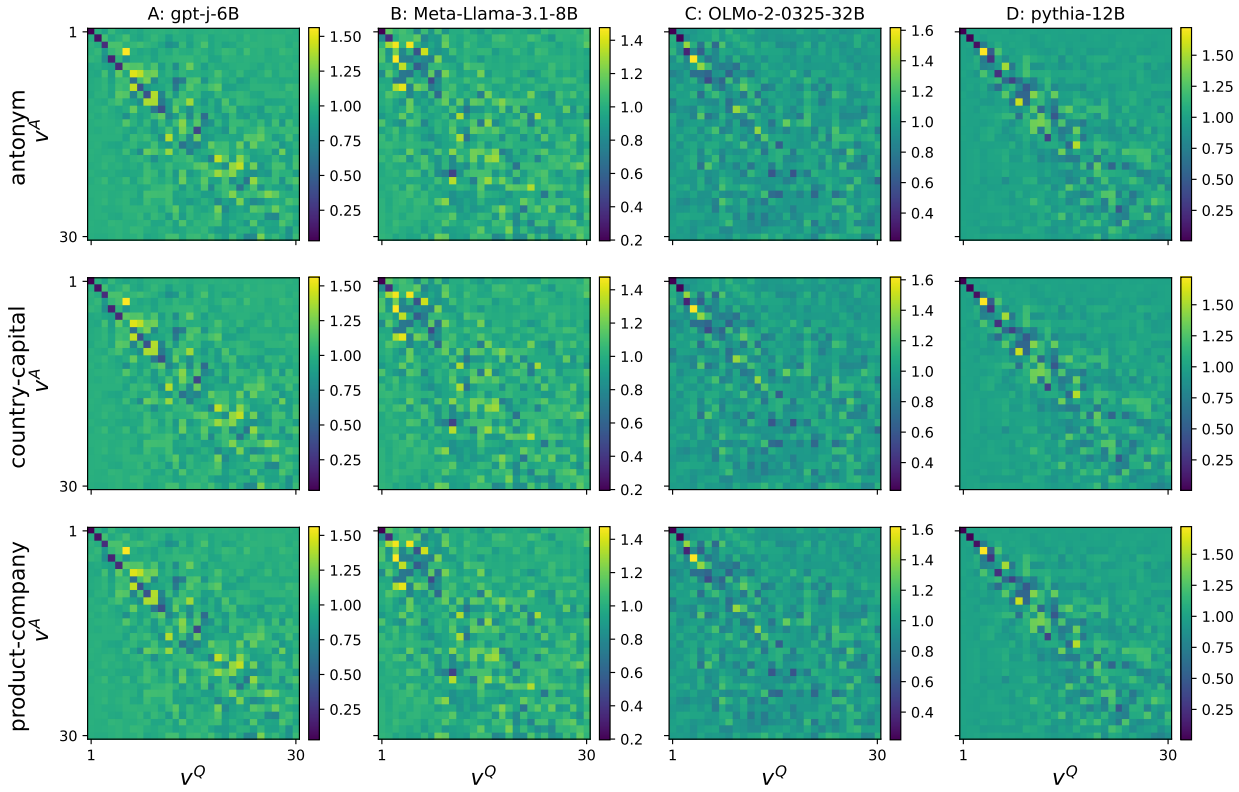

Figure 1: Cosine Distance between query and answer tokens. We computed the principal components ($v^A s$, $v^S s$, $v^Q s$) from answer (A), separator (S) and query (Q) tokens. Panels (A)-(D) show the cosine distance calculated from gpt-j-6B, Meta-Llama-3.1-8B, OLMo-2-0325-32B and Pythia-12B. The top, middle and bottom rows denote 'antonym', 'country-capital' and 'product-company'.

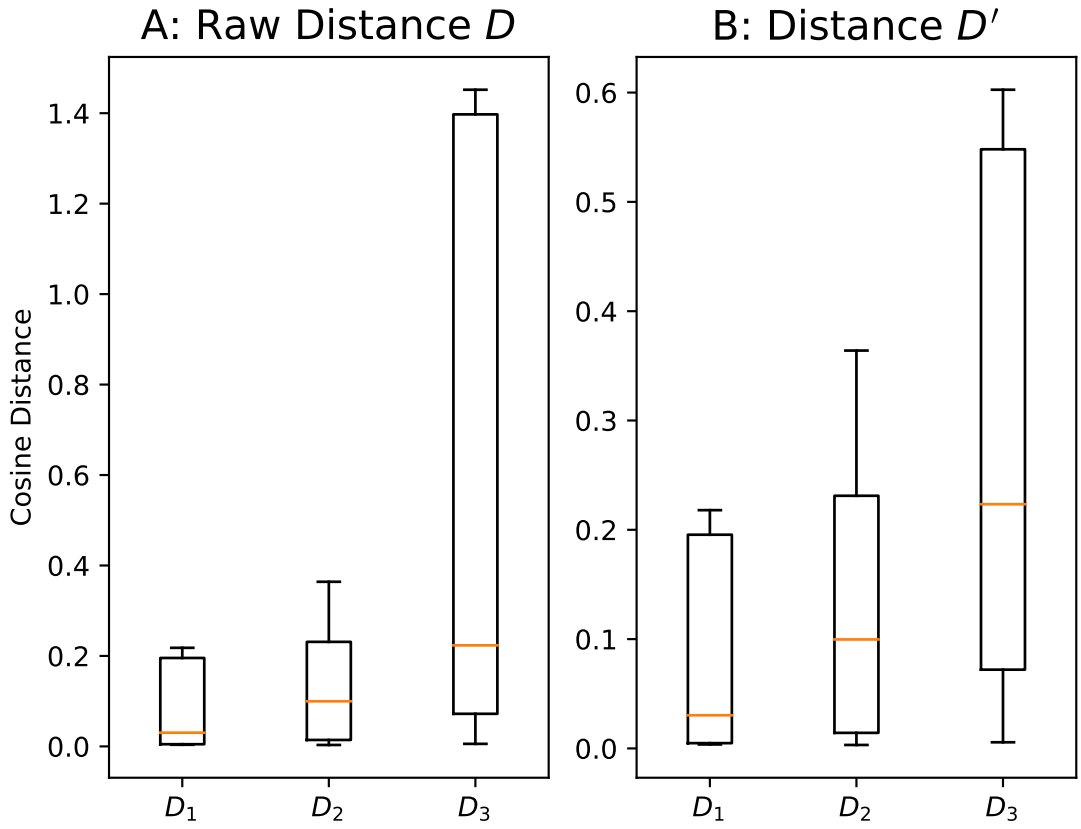

Figure 2: Distributions of distances between the first three principal components. (A), raw cosine distance D between $\vec{v}_1^Q$ and $\vec{v}_1^A$, between $\vec{v}_2^Q$ and $\vec{v}_2^A$ and between $\vec{v}_3^Q$ and $\vec{v}_3^A$. (B), the same as (A), but D' is displayed.

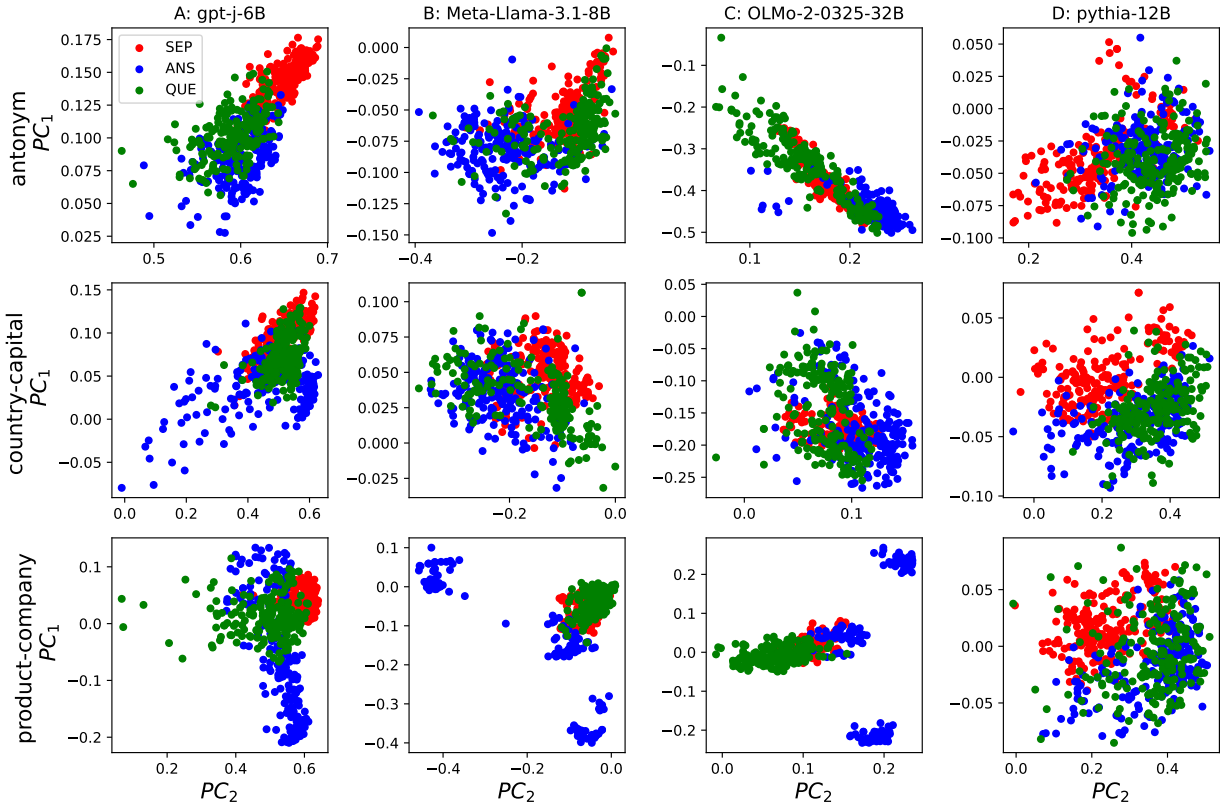

Figure 3: Tokens in the subspace spanned by v_1^A and v_2^A. We project answer (ANS), separator (SEP) and query (QUE) tokens to the subspace.(A)-(D) show the cosine distance calculated from gpt-j-6B, Meta-Llama-3.1-8B, OLMo-2-0325-32B and Pythia-12B. The top, middle and bottom rows denote 'antonym', 'country-capital' and 'product-company'